MIL-P-48232A(PA)
4 March 1974
SUPERSEDING
MIL-P-48232 (MU)
13 July 1973

MILITARY SPECIFICATION

PROPELLING CHARGE ASSEMBLY FOR ROCKETS, 3.5 INCH: HEAT M28A2, PRACTICE M29A2, AND WP M30

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the manufacture of one type of propellant grain for Propelling Charge For Use in 3.5 Inch Rocket.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-P-14737	-	Propellant, M7

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-652	-	Propellants, Solid, for Cannons; Requirements and Packing
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single or Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1340

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DRAWINGS

ORDNANCE CORPS

76-4-59	-	Box, Packing, Metal-Wood, M18 for Smokeless Powder Assembly
82-6-23	-	Rocket, Practice, 3.5 Inch M29A2, Assembly
82-16-35	-	Rocket HE, AT, 3.5 Inch M28A2, Practice M29A2 and Smoke, WP, M30 - Motor Loading Assembly
8858848	-	Marking Diagram and Sealing of Metal-Lined Wooden Packing Boxes for Shipment of Propellant

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Composition of Propellant. Composition of propellant shall be in accordance with MIL-P-14737 (MU) for Propellant, Type I.

3.3 Form. The form of the grain shall be cylindrical with a single axial perforation.

3.4 External Defects

3.4.1 The grains shall be free of visible surface defects (cracks or holes) exceeding 1/64 inch in the maximum external dimension. Die marks, screen marks, and marks made by grains touching during drying shall not be cause for rejection.

3.4.2 The grains shall be free of dirt, oil, grease and other foreign matter.

3.4.3 The ends of the grains shall be free of breaks or chips which extend either radially or longitudinally from the edge greater than one half the web thickness.

3.5 Grain Dimensions. The dimensions of the grain at 70 plus or minus (±) 5 degrees Fahrenheit (°F) shall be as follows:

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3.5.1 Length. 5.05 inches, maximum (max.).

3.5.2 Outer Diameter. 0.375 inch (advisory).

3.5.3 Web. 0.130 inch (advisory).

3.6 Straightness and Outer Diameter. The grains shall be straight enough and of proper size to be readily loaded into any acceptable combination of motor metal parts in accordance with Drawing (dwg.) 82-16-35. Any charge which fails to load freely or must be forced will be classified as defective. It is permissible to rotate a charge to permit the charge to load.

3.7 Charge Weight. A charge shall consist of 12 grains weighing within ± 1 1/2 grams of the lot mean established charge weight. The mean established charge weight shall be of such value as to produce the ballistics required in 3.8.

3.8 Ballistic Functioning. Propellant charges shall meet the following ballistic requirements:

3.8.1 Static Requirements

3.8.1.1 Maximum Pressure, 130°F. The mean maximum pressure (See 6.6) shall not exceed 14,000 psi. The maximum pressure of any individual charge shall not exceed 15,500 psi.

3.8.1.2 Burning Time, -20°F. The mean ten percent burning time (See 6.7) shall not exceed 0.045 second.

3.8.2 Flight Requirements, 70°F. The mean velocity (See 6.8) shall not exceed 328 feet per second (fps) and shall not be less than 320 fps. The standard deviation of the velocity shall not exceed 5 fps.

3.9 First Article Inspection. This specification does not make provisions for a first article inspection.

3.10 Workmanship. The parts and assemblies shall be free f rom chips, dirt, grease, rust and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents used.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Quality Conformance Inspection

4.2.1 Lot Formation. The term "inspection lot", as used throughout this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable subparagraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100%. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects, using an AQL of 0.40% for each Major Defect and an AQL of 0.65% for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraphs of this section, except where 100% inspection is specified, and be documented as part of the contractor's inspection system.

4.2.2.1 Propelling Charge (See dwg. 82-16-35C, covering a detail of dwg. 82-16-35).

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Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Total length, max.	Gage	01001
102.	Straightness	Gage	01002
103.	Outer diameter, (max.) of charge	Gage	01003
104.	Evidence of pin holes	Visual	01004
105.	Evidence of cracks	Visual	01005
Minor:	AQL 1.00 Percent		
201.	Evidence of chipped ends	Visual	01006
202.	Evidence of poor workmanship (See 3.10)	Visual	01007

4.2.3 Testing

4.2.3.1 Charge Weight (See 3.7) Major Defect, Code No. 01008. A sample of sixty (60) charges shall be selected from each lot and subjected to the test specified in 4.3.1. The lot shall be rejected if the heaviest charge is more than three (3) grams heavier than the lightest charge.

4.2.3.2 Static Test

4.2.3.2.1 Burning Time, -20°F (See 3.8.1.2) Major Defect, Code No. 01009. A sample of twenty (20) charges selected from those charges tested in 4.2.3.1 shall be subjected to the test specified in 4.3.2. The lot shall be rejected if the mean ten percent burning time exceeds 0.045 second.

4.2.3.2.2 Maximum Pressure, 130°F (See 3.8.1.1) Major Defects; Code No. 01010; Mean Maximum Pressure, Code No. 01011; Maximum Pressure. A sample of twenty (20) charges selected from those charges tested in 4.2.3.1 shall be subjected to the test specified in 4.3.2. The lot shall be rejected if any individual maximum pressure exceeds 15,500 psi or if the mean maximum pressure exceeds 14,000 psi.

4.2.3.3 Ballistic, 70°F (See 3.8.2) Major Defects, Code No. 01012; Mean Velocity, Code No. 01013; Velocity Standard Deviation. A sample of twenty (20) charges selected from those charges tested in 4.2.3.1 shall be subjected to the test specified in 4.3.3. The lot shall be rejected if the mean velocity is less than 320 fps or more than 328 fps or if the velocity standard deviation exceeds 5 fps.

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4.2.4 Inspection Equipment. The inspection equipment required to perform the examinations and tests prescribed in this section is identified in 4.2.3 (See 6.3).

4.3 Test Methods and Procedures. The test methods and procedures described in each of the following tests represent only one method and one procedure which has been proven in use and is approved by the Government for Government acceptance purposes. Producers and manufacturers have the option of using substitute methods and procedures only after obtaining Government approval. Government approval will be based on an evaluation to determine that the substitute method and procedure will result in an equal or better quality evaluation than that specified herein.

4.3.1 Charge Weight. The charge shall be weighed to the nearest 0.1 gram.

4.3.2 Static Tests. The round shall be loaded in accordance with dwg. 82-16-35, except that a test adaptor shall be provided to accommodate two (2) electric strain gages. The loaded test round shall be conditioned at the indicated temperature $\pm$ 2°F, for not less than 18 hours. If the conditioning chamber is not sufficiently near the test apparatus to permit the rounds to be tested within three minutes upon removal from the chamber, an insulated container having the same temperature within $\pm$ 2°F as the conditioning room shall be used to transport the rounds to the test apparatus. In such case, the rounds shall be stored in the insulated container, transported to the test apparatus and tested within three minutes after removal from the containers. Two pressure-time curves shall be obtained from each round, using the electric strain gages. Records in which only one pressure-time curve is legible shall not exceed 10 percent of the rounds fired at a given temperature. All rounds "lost" due to equipment failure shall be replaced by firing an equal number of additional rounds. Maximum pressure and 10 percent burning time for each round shall be measured and recorded. The mean of the readings obtained from the two strain gages shall be the official value of that variable for that round.

4.3.3 Ballistic Test. The round shall be loaded in accordance with dwg. 82-6-23, except that the weight of the complete round, including propellant, shall be 8.89 $\pm$ 0.02 pounds (See 6.10). The loaded test round shall be conditioned at the indicated temperature $\pm$ 2°F, for not less than 18 hours. If the conditioning chamber is not sufficiently near the test apparatus to permit the rounds to be tested within three minutes upon removal from the chamber, an insulated container having the same temperature

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within ± 2°F as the conditioning room shall be used to transport the rounds to the test apparatus. In such case, the rounds shall be stored in the insulated container, transported to the test apparatus and tested within three (3) minutes after removal from the containers. The round shall be fired from a standard 3.5 inch launcher of the latest design. The velocity of the round at a point 10 feet from the muzzle of the launcher, shall be determined by means of a chronograph or ribbon frame camera. When the velocity is determined by chronograph, the coils shall be spaced 10 feet ± .1 inch apart, with the first coil being located 5 feet from the forward edge of the launcher. The chronograph shall be read to the nearest millisecond.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Level C. Each charge consisting of 12 propellant grain selected in accordance with specification requirements shall be banded together with a rubber or neoprene band of sufficient strength to maintain the charge integrity.

5.2 Packing. Level C. Charges shall be packed in a horizontal position, within an M18 propellant container (dwg. 76-4-59). Net weight of charges per container shall be approximately 50 pounds. Cushioning consisting of PPP-P-291, Type III, Style 2 material shall be added when necessary, to fill container voids. Boxes shall be closed and sealed in accordance with instructions contained on dwg. C8858848.

5.3 Marking. Boxes shall be marked in accordance with dwg. C8858848 requirements.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.2 Inspection Code Numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on the Equipment Lists with the inspections listed in this specification.

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They should also be cited as references on drawings of equipment designs submitted by the contractor (See 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3, should he elect to do so.

6.3.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used

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with commerical equipment, or with SIE designs if not otherwise covered thereby (See 6.3.1 c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Visual Examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.5 Submission of Test Data and Ammunition Data Cards.-In addition to the normal distribution of records, when the assembly is procured by the Army Armaments Command, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-R and
SARPA-AD-E-P-1
Dover, New Jersey 07801

6.6 "Mean Maximum Pressure" is defined as the average of the maximum pressure obtained for each charge within the group and is expressed in pounds per square inch.

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6.7 "Ten Percent Burning Time" is defined as that time during which the pressure in the rocket is greater than 10 percent of its maximum value.

6.8 "Mean Velocity" is defined as the average of the velocity obtained for each charge within the group and is expressed in feet per second.

6.9 "Standard Deviation" computed from the sample is defined as the square root of the quotient of the sum of the square of the deviations of measurements of individual units from the sample mean and one less than the number of units in the sample. Any recognized method of computing standard deviation given an equivalent value may be used for calculation.

6.10 To provide rounds within the specified weight, inert material may be added in the motor closure cavity adjacent to the fuze to increase the weight; or the end of the head may be machined to decrease the weight.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1340-A405

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